

Initiating Search

November 10, 2025, 10:12 PM

• Search:

CAS Lexicon Search:

Pictet-Spengler cyclocondensation reaction - Preferred Concept

Search Tasks

Task	Result Type	View
Returned Reference Results from CAS Lexicon (1,783)	 References	View Results
Exported: Retrieved Related Reaction Results + Filters (486)	 Reactions	View Results
Filtered By:		
Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

Copyright © 2025 American Chemical Society (ACS). All Rights Reserved.

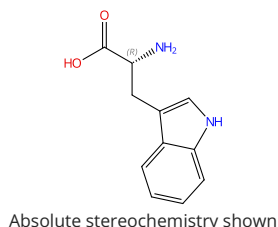
Internal use only. Redistribution is subject to the terms of your CAS SciFinder License Agreement and CAS information Use Policies.

Reactions (50)

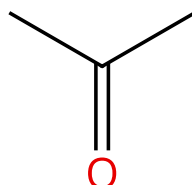
[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

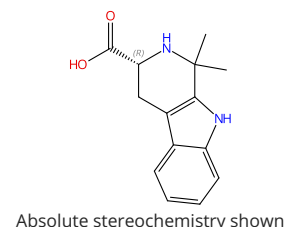
Steps: 1 Yield: 100%


 Suppliers (89)

+


 Suppliers (296)

→


 Supplier (1)

31-087-CAS-21351534

Steps: 1 Yield: 100%

Characterization of spatially addressable libraries: stereo isomer analysis of tetrahydro-β-carbolines as an example

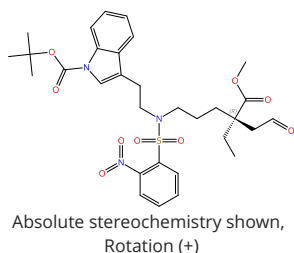
By: Cheng, Cesar C.; et al

Journal of Combinatorial Chemistry (1999), 1(6), 461-466.

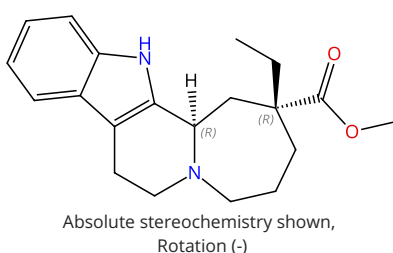
- 1.1 **Catalysts:** Trifluoroacetic acid
Solvents: Dichloromethane; 24 - 36 h, rt

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%



→



31-087-CAS-15760622

Steps: 1 Yield: 100%

Asymmetric Total Synthesis of Kopsiyunnanine K, a Monoterpenoid Indole Alkaloid with a Rearranged Skeleton

By: Tokuda, Ryoko; et al

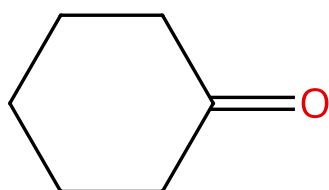
Organic Letters (2016), 18(14), 3490-3493.

- 1.1 **Reagents:** Thiophenol, Cesium carbonate
Solvents: Acetonitrile; 1.5 h, rt
- 1.2 **Reagents:** Water; rt
- 1.3 **Reagents:** Trifluoroacetic acid
Solvents: Dichloromethane; -30 °C → rt; 3 d, rt
- 1.4 **Reagents:** Sodium bicarbonate
Solvents: Water; rt

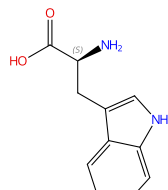
Experimental Protocols

Scheme 3 (1 Reaction)

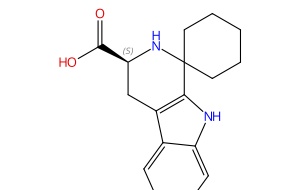
Steps: 1 Yield: 99%


 Suppliers (102)

+


 Suppliers (166)

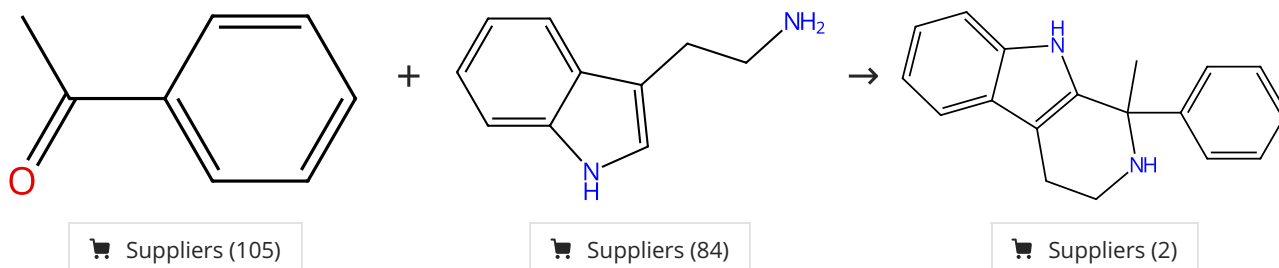
→


 Suppliers (3)

31-087-CAS-11037140 Steps: 1 Yield: 99% 1.1 Reagents: Trifluoroacetic acid Solvents: Dichloromethane; 10 min, 60 °C	Microwave accelerated Pictet-Spengler reactions of tryptophan with ketones directed toward the preparation of 1,1-disubstituted indole alkaloids By: Kuo, Fu-Ming; et al Tetrahedron (2004), 60(52), 12075-12084.
--	--

Scheme 4 (1 Reaction)

Steps: 1 Yield: 99%



31-087-CAS-361774 Steps: 1 Yield: 99% 1.1 Reagents: Titanium isopropoxide; 3 h, 80 °C 1.2 Catalysts: Trifluoroacetic acid Solvents: Trifluoroacetic anhydride; 3 h, 70 °C	A facile synthesis of 1,1-disubstituted 1,2,3,4-tetrahydro-β-carbolines via trifluoroacetic acid catalyzed Pictet-Spengler reaction using titanium(IV) isopropoxide as an imination reagent By: Horiguchi, Yoshie; et al Heterocycles (2003), 59(2), 691-705.
---	--

Scheme 5 (1 Reaction)

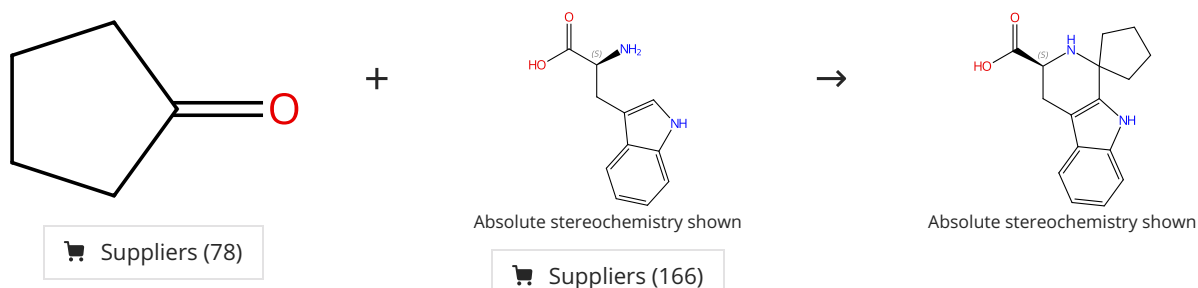
Steps: 1 Yield: 99%



31-087-CAS-21640128 Steps: 1 Yield: 99% 1.1 Reagents: Phosphorus oxychloride Solvents: Acetonitrile; 3 h, 80 °C Experimental Protocols	Synthesis of trans 8-substituted-6-phenyl-6,7,8,9-tetrahydro-3 H-pyrazolo[4,3-f]isoquinolines using a Pictet-Spengler approach By: Bailey, Andrew; et al Tetrahedron Letters (2018), 59(51), 4509-4513.
--	--

Scheme 6 (1 Reaction)

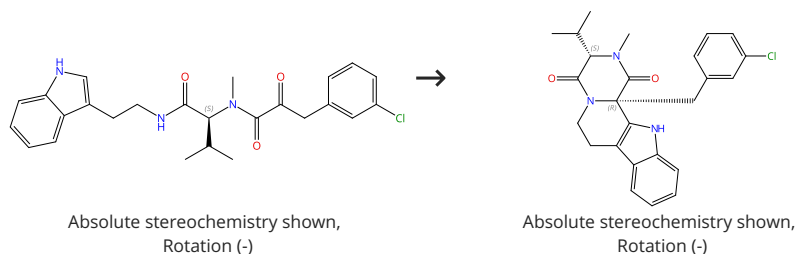
Steps: 1 Yield: 99%



31-087-CAS-1907448	Steps: 1 Yield: 99%	Microwave accelerated Pictet-Spengler reactions of tryptophan with ketones directed toward the preparation of 1,1-disubstituted indole alkaloids By: Kuo, Fu-Ming; et al Tetrahedron (2004), 60(52), 12075-12084.
1.1 Reagents: Trifluoroacetic acid Solvents: Dichloromethane; 20 min, 60 °C		

Scheme 7 (1 Reaction)

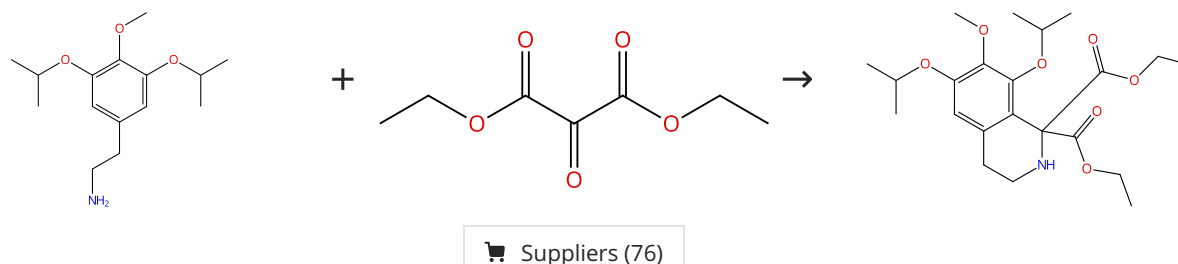
Steps: 1 Yield: 99%



31-087-CAS-9417929	Steps: 1 Yield: 99%	Diastereodivergent synthesis of 2,5-diketopiperazine derivatives of β-carboline and isoquinoline from L-amino acids By: Siwicka, Aleksandra; et al Tetrahedron: Asymmetry (2005), 16(5), 975-993.
1.1 Reagents: Hydrochloric acid Solvents: Ethyl acetate; overnight, rt		

Scheme 8 (1 Reaction)

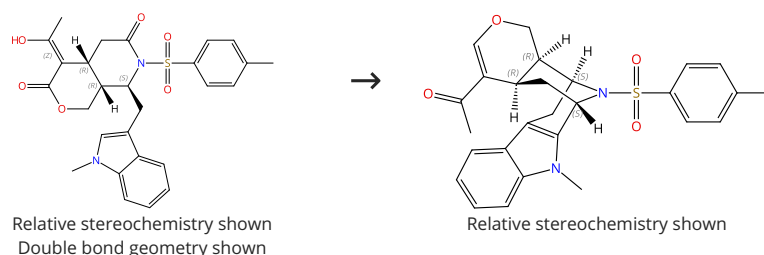
Steps: 1 Yield: 98%



31-087-CAS-14198792	Steps: 1 Yield: 98%	Diethyl oxomalonate as a three carbon synthon for synthesis of functionalized 1,1'-disubstituted tetrahydroisoquinoline By: Bois-Choussy, M.; et al Tetrahedron Letters (2001), 42(27), 4503-4506.
1.1 Reagents: Trifluoroacetic acid Solvents: Toluene		

Scheme 9 (1 Reaction)

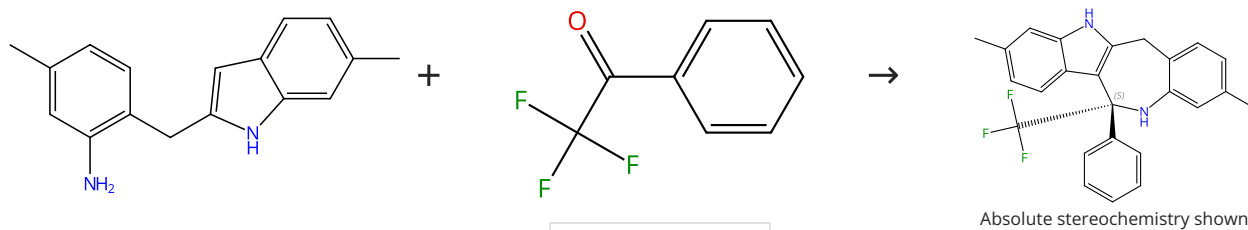
Steps: 1 Yield: 98%



31-087-CAS-1965039	Steps: 1 Yield: 98%	Aziridine-based concise synthesis of ($\pm$)-alstonerine By: Craig, Donald; et al Chemical Communications (Cambridge, United Kingdom) (2013), 49(81), 9275-9277.
1.1 Reagents: Diisobutylaluminum hydride Solvents: Dichloromethane, Toluene; 2.25 h, -78 °C 1.2 Reagents: Diethyl ether, Water; -78 °C $\rightarrow$ rt 1.3 Reagents: Dichloromethane, Water; 1 h, rt Experimental Protocols		

Scheme 10 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (85)

31-087-CAS-7541385

Steps: 1 Yield: 98%

- 1.1 **Catalysts:** Diindeno[7,1-*de*:1',7'-*fg*][1,3,2]dioxaphosphocin, 3, 7-bis[3,5-bis(trifluoromethyl)phenyl]-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11a*S*)-
Solvents: 1,2-Dichloroethane; 24 - 96 h, 50 °C

Experimental Protocols

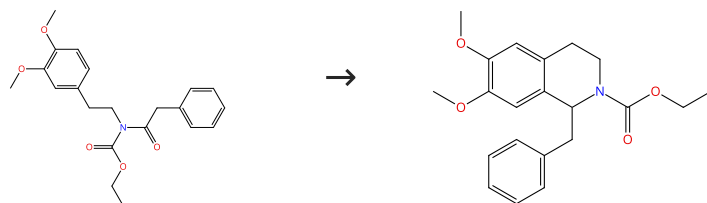
Enantioselective synthesis of benzazepinoindoles bearing trifluoromethylated quaternary stereocenters catalyzed by chiral spirocyclic phosphoric acids

By: Li, Xuejian; et al

Chemical Communications (Cambridge, United Kingdom) (2014), 50(56), 7538-7541.

Scheme 11 (1 Reaction)

Steps: 1 Yield: 98%



31-087-CAS-9457442

Steps: 1 Yield: 98%

- 1.1 **Reagents:** Diisobutylaluminum hydride
Solvents: Dichloromethane; -78 °C; 1 h, -78 °C
 1.2 **Reagents:** Boron trifluoride etherate; 1 h, -78 °C
 1.3 **Reagents:** Potassium sodium tartrate

Experimental Protocols

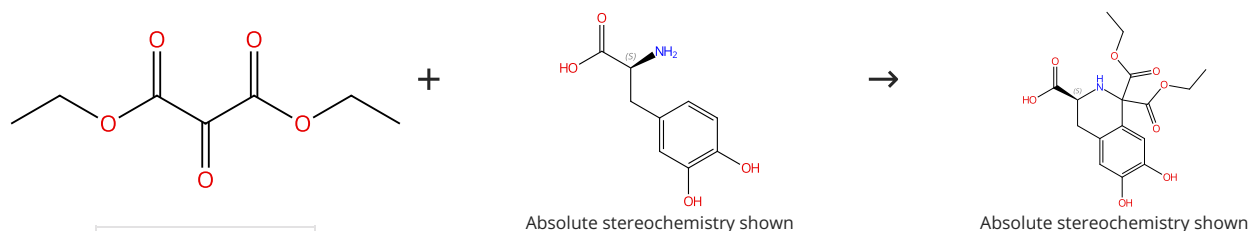
A variation of the Pictet-Spengler reaction via a sequential reduction-cyclization reaction of N-acylcarbamates: Synthesis of 1-substituted tetrahydroisoquinoline derivatives

By: Kuhakarn, Chutima; et al

Tetrahedron Letters (2007), 48(46), 8182-8184.

Scheme 12 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (76)

Suppliers (139)

31-087-CAS-7206086

Steps: 1 Yield: 98%

- 1.1 **Reagents:** Trifluoroacetic acid
Solvents: Toluene

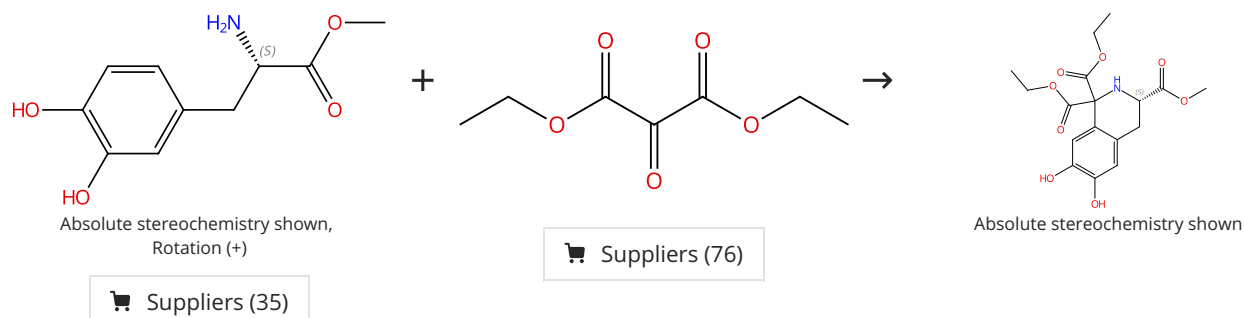
Diethyl oxomalonate as a three carbon synthon for synthesis of functionalized 1,1'-disubstituted tetrahydroisoquinoline

By: Bois-Choussy, M.; et al

Tetrahedron Letters (2001), 42(27), 4503-4506.

Scheme 13 (1 Reaction)

Steps: 1 Yield: 98%



31-087-CAS-3218048

Steps: 1 Yield: 98%

Diethyl oxomalonate as a three carbon synthon for synthesis of functionalized 1,1'-disubstituted tetrahydroisoquinoline

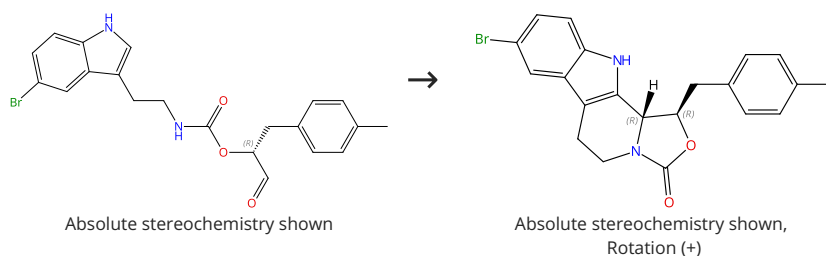
1.1 Reagents: Trifluoroacetic acid
Solvents: Toluene

By: Bois-Choussy, M.; et al

Tetrahedron Letters (2001), 42(27), 4503-4506.

Scheme 14 (1 Reaction)

Steps: 1 Yield: 98%



31-087-CAS-11511717

Steps: 1 Yield: 98%

Asymmetric total synthesis of reported structure of eudistomidin B, an indole alkaloid isolated from a marine tunicate

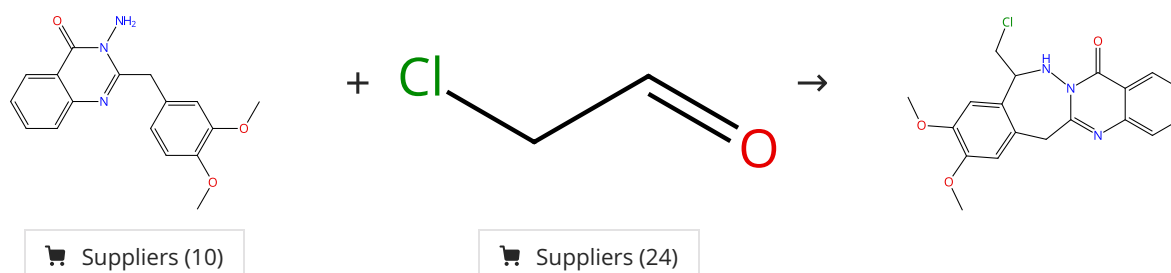
1.1 Reagents: Trifluoroacetic acid
Solvents: Dichloromethane; 0 °C

By: Ito, Takeshi; et al

Tetrahedron Letters (2009), 50(31), 4506-4508.

Scheme 15 (1 Reaction)

Steps: 1 Yield: 98%



31-087-CAS-10088589

Steps: 1 Yield: 98%

Pictet-Spengler reaction in the synthesis of condensed benzodiazepines. Part 2. Synthesis of new derivatives of 11, 12-dihydroquinazoline[3,2-c][2,3]benzodiazepin-14(6H)-ones

1.1 Reagents: Hydrochloric acid
Solvents: 1,4-Dioxane, Water; 1 h, 80 - 90 °C; 80 °C → rt

By: Tolkunov, A. S.; et al

1.2 Reagents: Ammonium hydroxide
Solvents: Water; 1 h, pH 4 - 5, rt

Chemistry of Heterocyclic Compounds (New York, NY, United States) (2010), 46(6), 711-720.

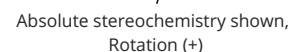
Experimental Protocols

Steps: **1** Yield: **97%**

Biomimetic Phosphate-Catalyzed Pictet-Spengler Reaction for the Synthesis of 1,1'-Disubstituted and Spiro-Tetrahydroisoquinoline Alkaloids

Journal of Organic Chemistry (2019), 84(12), 7702-7710.

Steps: **1** Yield: **97%**



An Easy Entry to Optically Active Spiroindolinones: Chiral Bronsted Acid-Catalysed Pictet-Spengler Reactions of Isatins

Advanced Synthesis & Catalysis (2011), 353(6), 860-864.

Experimental Protocols

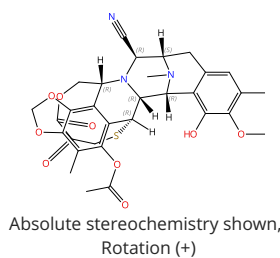
Steps: **1** Yield: **97%**

Biomimetic Phosphate-Catalyzed Pictet-Spengler Reaction for the Synthesis of 1,1'-Disubstituted and Spiro-Tetrahydroisoquinoline Alkaloids

Journal of Organic Chemistry (2019), 84(12), 7702-7710.

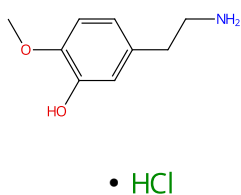
Scheme 19 (1 Reaction)

Steps: 1 Yield: 97%



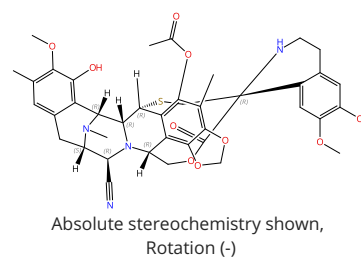
Suppliers (13)

+



Suppliers (63)

→



Suppliers (37)

31-087-CAS-2444256

Steps: 1 Yield: 97%

Total Synthesis of Ecteinasidin 743

1.1 Reagents: Sodium acetate
Solvents: Ethanol; 3 h, 23 °C

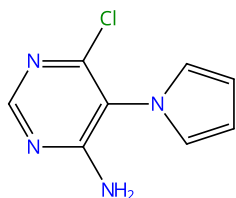
Experimental Protocols

By: Chen, Jinchun; et al

Journal of the American Chemical Society (2006), 128(1), 87-89.

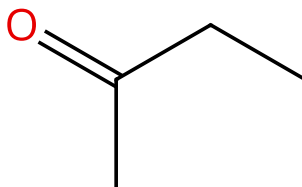
Scheme 20 (1 Reaction)

Steps: 1 Yield: 96%



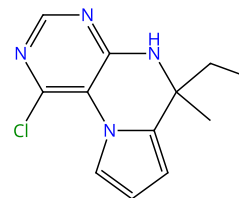
Suppliers (4)

+



Suppliers (116)

→



31-087-CAS-3168105

Steps: 1 Yield: 96%

Synthesis of novel 1,6-disubstituted 5,6-dihydropyrrolo[1,2-f]pteridines

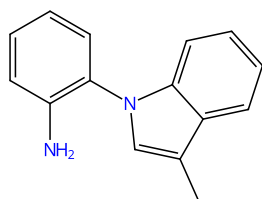
1.1 Catalysts: *p*-Toluenesulfonic acid
Solvents: Toluene; rt → reflux; reflux; reflux → rt

By: Zheng, Lian-You; et al

Gaodeng Xuexiao Huaxue Xuebao (2006), 27(10), 1869-1872.

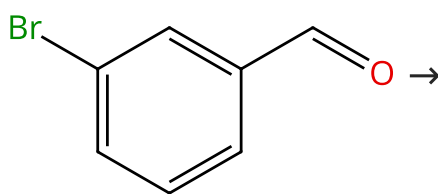
Scheme 21 (1 Reaction)

Steps: 1 Yield: 96%



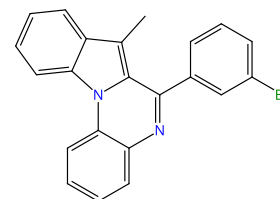
Suppliers (2)

+



Suppliers (91)

→



31-087-CAS-13005562

Steps: 1 Yield: 96%

Lewis Acid-Catalyzed Selective Synthesis of Diversely Substituted Indolo- and Pyrrolo[1,2-a]quinoxalines and Quinoxalinones by Modified Pictet-Spengler Reaction

1.1 Reagents: 1*H*-Benzotriazole, Oxygen
Catalysts: Aluminum chloride
Solvents: Tetrahydrofuran; 8 - 10 h, rt

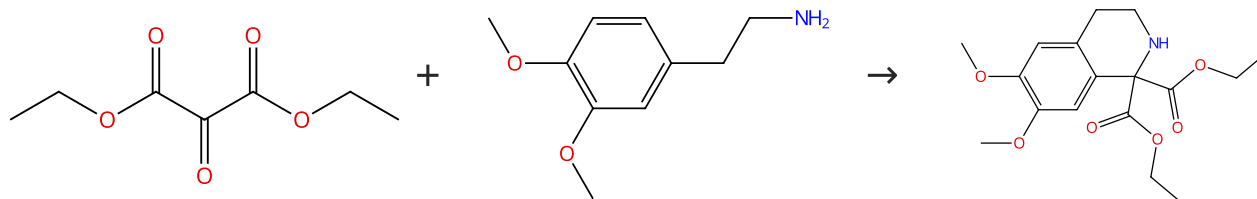
By: Verma, Akhilesh K.; et al

Experimental Protocols

European Journal of Organic Chemistry (2011), 2011(34), 6998-7010, S6998/1-S6998/109.

Scheme 22 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (76)

Suppliers (64)

31-087-CAS-7533498

Steps: 1 Yield: 96%

Diethyl oxomalonate as a three carbon synthon for synthesis of functionalized 1,1'-disubstituted tetrahydroisoquinoline

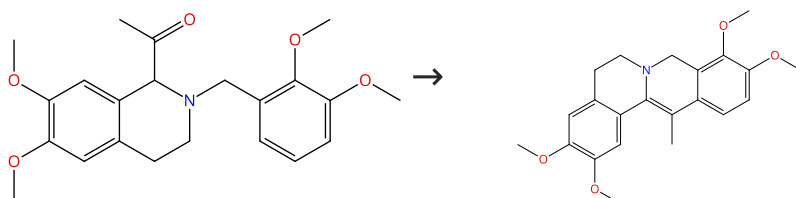
1.1 Reagents: Trifluoroacetic acid
Solvents: Toluene

By: Bois-Choussy, M.; et al

Tetrahedron Letters (2001), 42(27), 4503-4506.

Scheme 23 (1 Reaction)

Steps: 1 Yield: 96%



31-614-CAS-37225920

Steps: 1 Yield: 96%

An Efficient and Eco-Friendly Synthesis of Protoberberine and 13-Me Protoberberine Alkaloids via Electrochemical Accept orless Dehydrogenation

1.1 Reagents: Hydrochloric acid
Solvents: Water; 6 h, rt

1.2 Reagents: Sodium carbonate
Solvents: Water; pH 10, rt

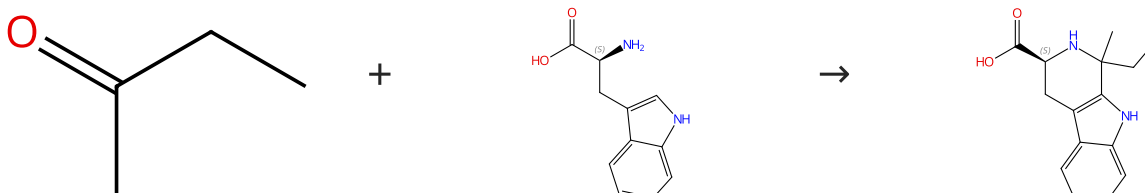
By: Li, Weijian; et al

ChemCatChem (2023), 15(7), e202201553.

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (116)

Absolute stereochemistry shown

Suppliers (166)

Absolute stereochemistry shown

Suppliers (4)

31-087-CAS-8622810

Steps: 1 Yield: 96%

Microwave accelerated Pictet-Spengler reactions of tryptophan with ketones directed toward the preparation of 1,1-disubstituted indole alkaloids

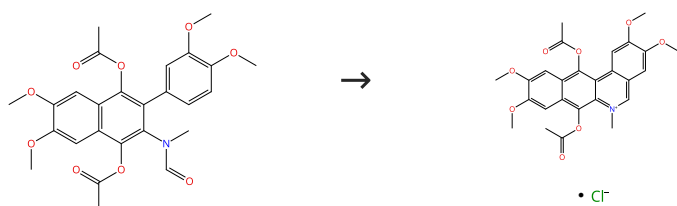
1.1 Reagents: Trifluoroacetic acid
Solvents: Dichloromethane; 20 min, 60 °C

By: Kuo, Fu-Ming; et al

Tetrahedron (2004), 60(52), 12075-12084.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 96%



31-087-CAS-1855277

Steps: 1 Yield: 96%

Pictet-Spengler and Bischler-Napieralski syntheses of benzo [b]phenanthridines

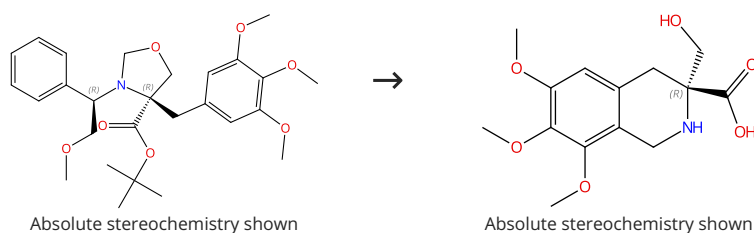
By: Gogilashvili, T. I.; et al

Khimiko-Farmatsevticheskii Zhurnal (1989), 23(12), 1436-40.

1.1 Reagents: Phosphorus oxychloride
Solvents: Acetonitrile

Scheme 26 (1 Reaction)

Steps: 1 Yield: 96%



31-087-CAS-12073748

Steps: 1 Yield: 96%

Asymmetric synthesis of quaternary tetrahydroisoquinoline-3-carboxylic acid derivatives

By: Alezra, V.; et al

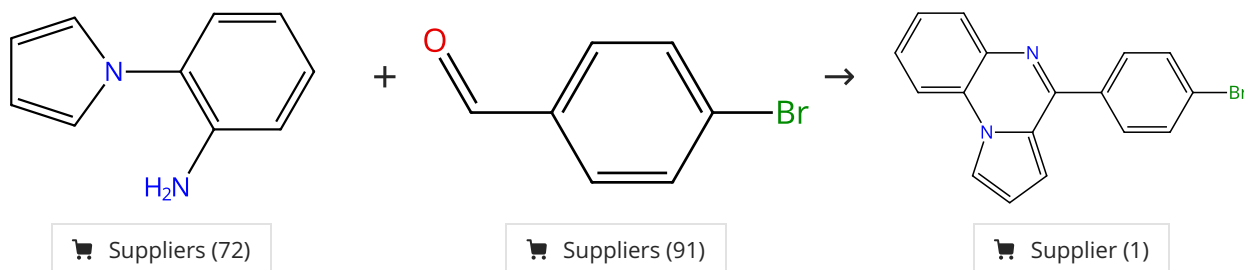
Tetrahedron Letters (2001), 42(11), 2111-2113.

1.1 Reagents: Titanium tetrachloride
Catalysts: Triethylamine
Solvents: Dichloromethane

1.2 Reagents: Hydrogen
Catalysts: Palladium dihydroxide
Solvents: Ethanol, Water

Scheme 27 (1 Reaction)

Steps: 1 Yield: 96%



31-087-CAS-6328127

Steps: 1 Yield: 96%

Lewis Acid-Catalyzed Selective Synthesis of Diversely Substituted Indolo- and Pyrrolo[1,2-a]quinoxalines and Quinoxalinones by Modified Pictet-Spengler Reaction

By: Verma, Akhilesh K.; et al

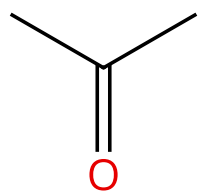
European Journal of Organic Chemistry (2011), 2011(34), 6998-7010, S6998/1-S6998/109.

1.1 Reagents: 1H-Benzotriazole, Oxygen
Catalysts: Aluminum chloride
Solvents: Tetrahydrofuran; 8 - 10 h, rt

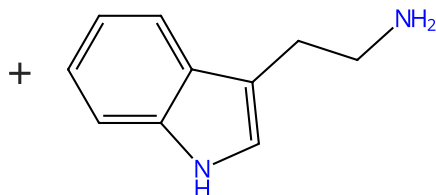
Experimental Protocols

Scheme 28 (1 Reaction)

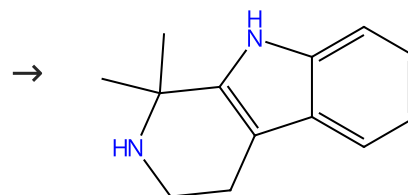
Steps: 1 Yield: 96%



Suppliers (296)



Suppliers (84)



Suppliers (6)

31-087-CAS-10841955

Steps: 1 Yield: 96%

- 1.1 Reagents: Sodium hydroxide
Solvents: Ethanol, Water; 18 h, reflux

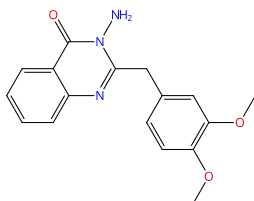
A facile synthesis of 1,1-disubstituted 1,2,3,4-tetrahydro-β-carbolines via trifluoroacetic acid catalyzed Pictet-Spengler reaction using titanium(IV) isopropoxide as an imination reagent

By: Horiguchi, Yoshie; et al

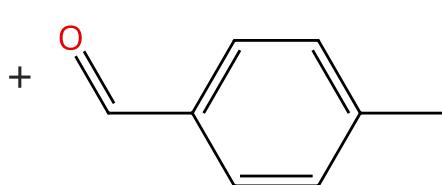
Heterocycles (2003), 59(2), 691-705.

Scheme 29 (1 Reaction)

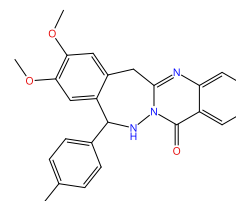
Steps: 1 Yield: 95%



Suppliers (10)



Suppliers (104)



Supplier (1)

31-087-CAS-13521318

Steps: 1 Yield: 95%

- 1.1 Reagents: Hydrochloric acid
Solvents: 1,4-Dioxane, Water; 2 - 3 h, 90 °C; cooled

- 1.2 Reagents: Ammonia
Solvents: Water; 1 h, basified

Experimental Protocols

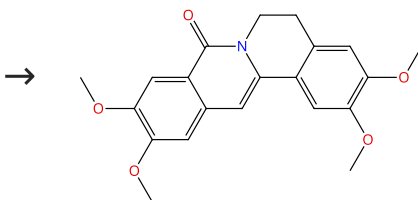
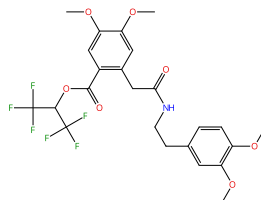
The Pictet-Spengler reaction in the synthesis of condensed benzodiazepines 1. synthesis of 6,11,12,14-tetrahydrobenzo-[4,5][1,2]diazepino[7,1-b]quinazolin-14-ones

By: Tolkunov, A. S.; et al

Chemistry of Heterocyclic Compounds (New York, NY, United States) (2010), 46(5), 592-599.

Scheme 30 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (9)

31-087-CAS-23637173

Steps: 1 Yield: 95%

- 1.1 Reagents: Phosphorus oxychloride
Solvents: Toluene; rt; rt → 115 °C; 115 °C → 80 °C

- 1.2 Reagents: Potassium carbonate
Catalysts: Tetrabutylammonium bromide
Solvents: Methanol; overnight, 80 °C

Experimental Protocols

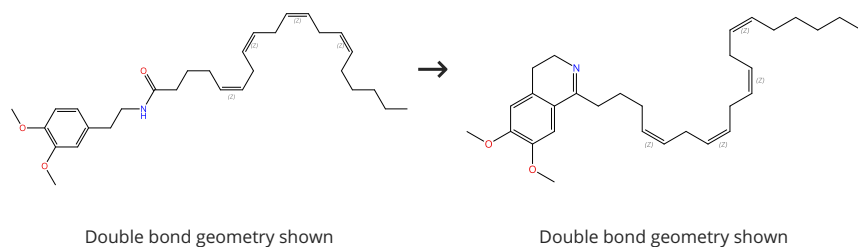
Unified synthesis of polycyclic alkaloids via complementary carbonyl activation

By: He, Guoli; et al

Angewandte Chemie, International Edition (2021), 60(24), 13591-13596.

Scheme 31 (1 Reaction)

Steps: 1 Yield: 95%



31-087-CAS-11716666

Steps: 1 Yield: 95%

Synthesis of a novel class of fatty acids-derived isoquinolines

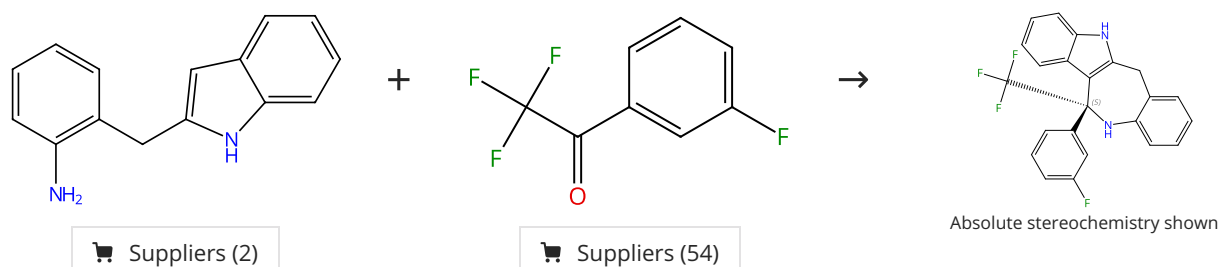
- 1.1 **Reagents:** Phosphorus pentachloride
Solvents: Dichloromethane; 20 min, 0 °C; 6 h, rt
- 1.2 **Reagents:** Sodium bicarbonate
Solvents: Water

By: Matuszewska, Iwona; et al

Chemistry and Physics of Lipids (2005), 135(2), 131-145.

Scheme 32 (1 Reaction)

Steps: 1 Yield: 95%



31-087-CAS-10158581

Steps: 1 Yield: 95%

Enantioselective synthesis of benzazepinoindoles bearing trifluoromethylated quaternary stereocenters catalyzed by chiral spirocyclic phosphoric acids

- 1.1 **Catalysts:** Diindeno[7,1-*de*:1',7'-*fg*][1,3,2]dioxaphosphocin, 3, 7-bis[3,5-bis(trifluoromethyl)phenyl]-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11*aS*)
Solvents: 1,2-Dichloroethane; 24 - 48 h, 35 °C

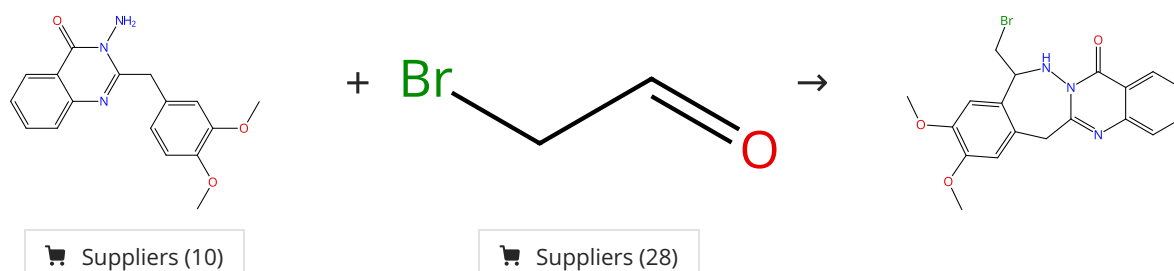
By: Li, Xuejian; et al

Chemical Communications (Cambridge, United Kingdom) (2014), 50(56), 7538-7541.

Experimental Protocols

Scheme 33 (1 Reaction)

Steps: 1 Yield: 95%



31-087-CAS-12224458

Steps: 1 Yield: 95%

Pictet-Spengler reaction in the synthesis of condensed benzodiazepines. Part 2. Synthesis of new derivatives of 11, 12-dihydroquinazoline[3,2-*c*][2,3]benzodiazepin-14(6*H*)-ones

- 1.1 **Reagents:** Hydrochloric acid
Solvents: 1,4-Dioxane, Water; 1 h, 80 - 90 °C; 80 °C → rt
- 1.2 **Reagents:** Ammonium hydroxide
Solvents: Water; 1 h, pH 4 - 5, rt

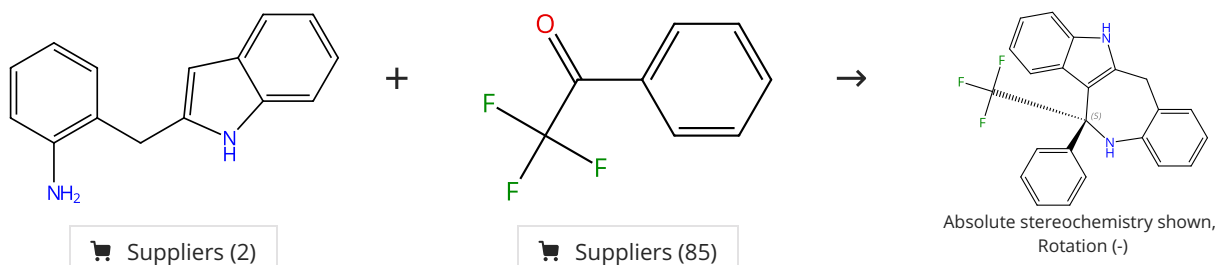
By: Tolkunov, A. S.; et al

Chemistry of Heterocyclic Compounds (New York, NY, United States) (2010), 46(6), 711-720.

Experimental Protocols

Scheme 34 (1 Reaction)

Steps: 1 Yield: 95%



31-087-CAS-14074910

Steps: 1 Yield: 95%

- 1.1 **Catalysts:** Diindeno[7,1-*de*:1',7'-*fg*][1,3,2]dioxaphosphocin, 3, 7-bis[3,5-bis(trifluoromethyl)phenyl]-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11a*S*)-
Solvents: 1,2-Dichloroethane; 24 h, 35 °C

Experimental Protocols

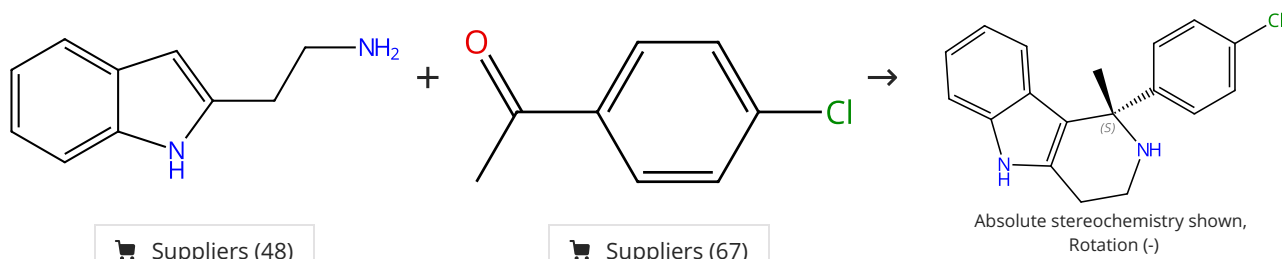
Enantioselective synthesis of benzazepinoindoles bearing trifluoromethylated quaternary stereocenters catalyzed by chiral spirocyclic phosphoric acids

By: Li, Xuejian; et al

Chemical Communications (Cambridge, United Kingdom) (2014), 50(56), 7538-7541.

Scheme 35 (1 Reaction)

Steps: 1 Yield: 95%



31-087-CAS-13768154

Steps: 1 Yield: 95%

- 1.1 **Reagents:** Benzoic acid
Catalysts: (2*S*)-2-[[[3,5-Bis(trifluoromethyl)phenyl]amino]thioxomethyl]amino]-*N*-(diphenylmethyl)-*N*,3,3-trimethylbutanamide
Solvents: Toluene; 1 h, 22 °C

Experimental Protocols

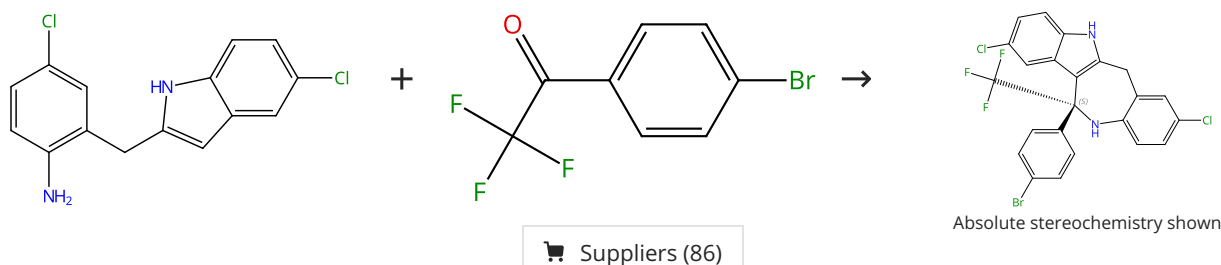
Thiourea-Catalyzed Enantioselective Iso-Pictet-Spengler Reactions

By: Lee, Yun-Mi; et al

Organic Letters (2011), 13(20), 5564-5567.

Scheme 36 (1 Reaction)

Steps: 1 Yield: 95%



31-087-CAS-12134393

Steps: 1 Yield: 95%

- 1.1 **Catalysts:** Diindeno[7,1-*de*:1',7'-*fg*][1,3,2]dioxaphosphocin, 3, 7-bis[3,5-bis(trifluoromethyl)phenyl]-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11a*S*)-
Solvents: 1,2-Dichloroethane; 24 - 96 h, reflux

Experimental Protocols

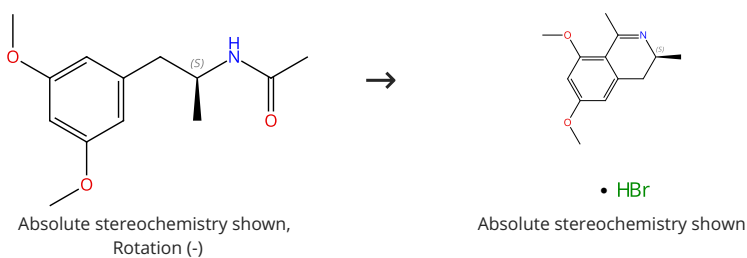
Enantioselective synthesis of benzazepinoindoles bearing trifluoromethylated quaternary stereocenters catalyzed by chiral spirocyclic phosphoric acids

By: Li, Xuejian; et al

Chemical Communications (Cambridge, United Kingdom) (2014), 50(56), 7538-7541.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 95%



31-087-CAS-12561637

Steps: 1 Yield: 95%

1.1 Reagents: Phosphorus oxychloride
Solvents: Acetonitrile

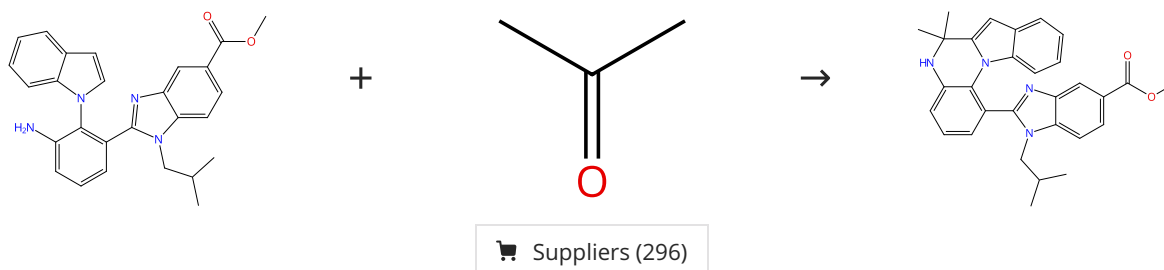
The synthesis of all possible isomeric 6,8-dioxygenated 1,3-dimethyl-1,2,3,4-tetrahydroisoquinoline methyl ethers - useful chiral building blocks for naphthylisoquinoline alkaloids

By: Bringmann, Gerhard; et al

Liebigs Annalen der Chemie (1993), (8), 877-88.

Scheme 38 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (296)

31-087-CAS-1852882

Steps: 1 Yield: 95%

1.1 Reagents: Zinc chloride
Catalysts: Trifluoroacetic acid
Solvents: Chloroform; 14 h, reflux

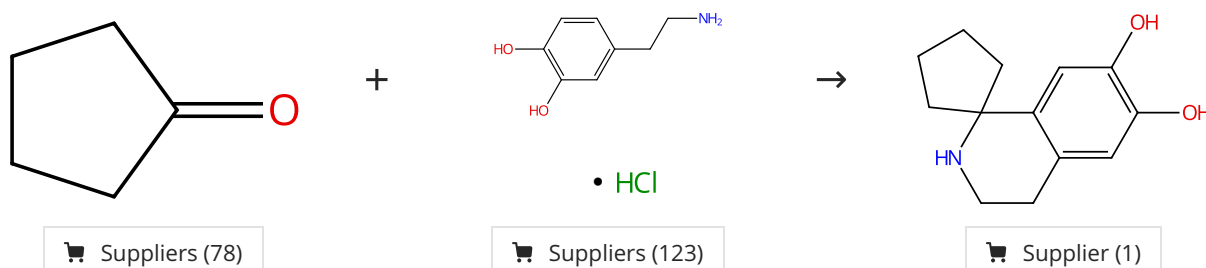
Design and Synthesis of New Biprivileged Molecular Scaffolds: Indolo-Fused Benzodiazepinyl/quinoxaliny benzimidazoles

By: Barve, Indrajeet J.; et al

Chemistry - An Asian Journal (2012), 7(7), 1684-1690.

Scheme 39 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (78)

Suppliers (123)

Supplier (1)

31-087-CAS-22095346

Steps: 1 Yield: 95%

1.1 Reagents: Sodium ascorbate
Solvents: Methanol, Water; 20 h, pH 9, 70 °C

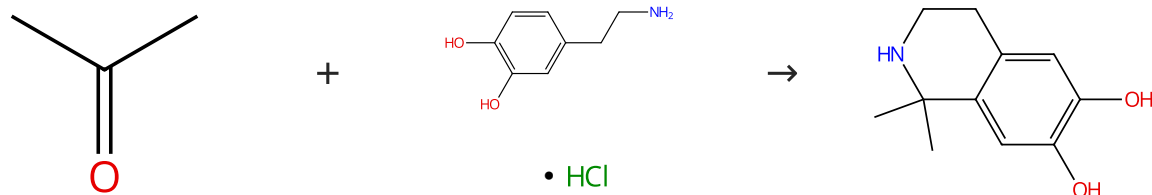
Biomimetic Phosphate-Catalyzed Pictet-Spengler Reaction for the Synthesis of 1,1'-Disubstituted and Spiro-Tetrahydroisoquinoline Alkaloids

By: Zhao, Jianxiong; et al

Journal of Organic Chemistry (2019), 84(12), 7702-7710.

Scheme 40 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (296)

Suppliers (123)

Suppliers (15)

31-087-CAS-21422802

Steps: 1 Yield: 95%

Biomimetic Phosphate-Catalyzed Pictet-Spengler Reaction for the Synthesis of 1,1'-Disubstituted and Spiro-Tetrahydroisoquinoline Alkaloids

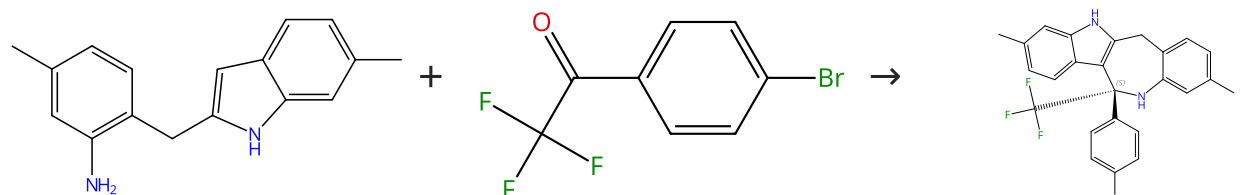
By: Zhao, Jianxiong; et al

Journal of Organic Chemistry (2019), 84(12), 7702-7710.

1.1 **Reagents:** Sodium ascorbate
Solvents: Methanol, Water; 20 h, pH 9, 70 °C

Scheme 41 (1 Reaction)

Steps: 1 Yield: 95%



Absolute stereochemistry shown

Suppliers (86)

31-087-CAS-4798741

Steps: 1 Yield: 95%

Enantioselective synthesis of benzazepinoindoles bearing trifluoromethylated quaternary stereocenters catalyzed by chiral spirocyclic phosphoric acids

By: Li, Xuejian; et al

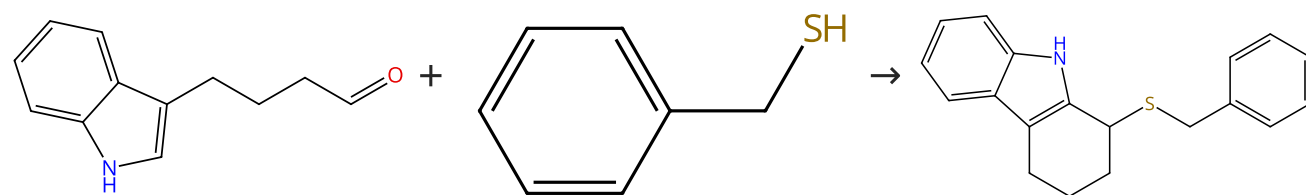
Chemical Communications (Cambridge, United Kingdom) (2014), 50(56), 7538-7541.

1.1 **Catalysts:** Diindeno[7,1-*de*:1',7'-*fg*][1,3,2]dioxaphosphocin, 3, 7-bis[3,5-bis(trifluoromethyl)phenyl]-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11a*S*)-
Solvents: 1,2-Dichloroethane; 24 - 96 h, 50 °C

Experimental Protocols

Scheme 42 (1 Reaction)

Steps: 1 Yield: 95%



Suppliers (17)

Suppliers (53)

31-087-CAS-16361603

Steps: 1 Yield: 95%

Catalytic Enantioselective Synthesis of Tetrahydrocarbazoles and Exocyclic Pictet-Spengler-Type Reactions

By: Hansen, Casper L.; et al

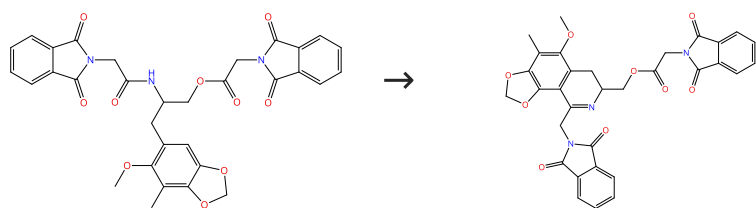
Organic Letters (2016), 18(23), 5990-5993.

1.1 **Catalysts:** Dinaphtho[2,1-*d*:1',2'-*f*][1,3,2]dioxaphosphepin, 4-hydroxy-2,6-bis[2,4,6-tris(1-methylethyl)phenyl]-, 4-oxide, (11b*R*)-
Solvents: Dichloromethane; 5 h, -50 °C

Experimental Protocols

Scheme 43 (1 Reaction)

Steps: 1 Yield: 95%



31-087-CAS-13847973

Steps: 1 Yield: 95%

Synthesis of (±)-phthalascidin 622

1.1 **Reagents:** Phosphorus oxychloride
Solvents: Acetonitrile; 0 °C; 3 h, 90 °C

1.2 **Reagents:** Sodium carbonate
Solvents: Water; neutralized, 0 °C

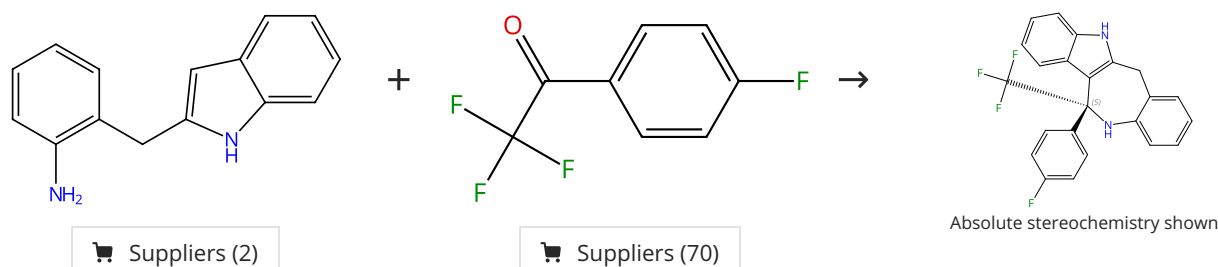
Experimental Protocols

By: Christian, R. Razafindrabe; et al

Science China: Chemistry (2010), 53(9), 1888-1898.

Scheme 44 (1 Reaction)

Steps: 1 Yield: 95%



31-087-CAS-8357958

Steps: 1 Yield: 95%

Enantioselective synthesis of benzazepinoindoles bearing trifluoromethylated quaternary stereocenters catalyzed by chiral spirocyclic phosphoric acids

1.1 **Catalysts:** Diindeno[7,1-*de*:1',7'-*fg*][1,3,2]dioxaphosphocin, 3, 7-bis[3,5-bis(trifluoromethyl)phenyl]-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11a*S*)
Solvents: 1,2-Dichloroethane; 24 - 48 h, 35 °C

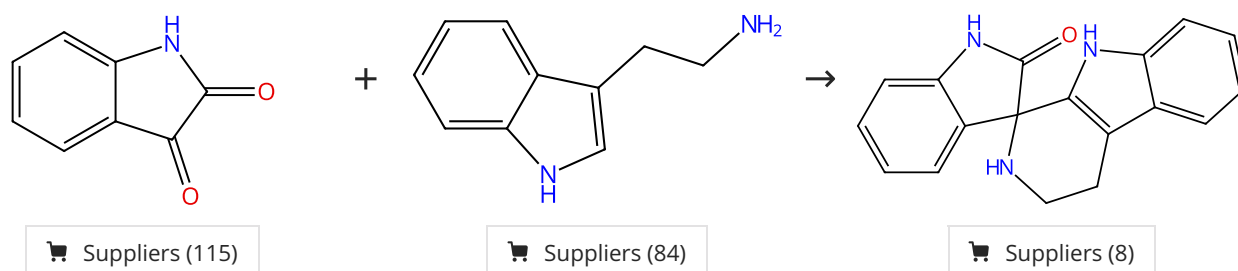
Experimental Protocols

By: Li, Xuejian; et al

Chemical Communications (Cambridge, United Kingdom) (2014), 50(56), 7538-7541.

Scheme 45 (1 Reaction)

Steps: 1 Yield: 95%



31-087-CAS-18104062

Steps: 1 Yield: 95%

Bioglycerol-derived carbon-SO₃H as a recyclable catalyst for the synthesis of tetrahydro-β-carbolines

1.1 **Catalysts:** Glycerol (bioglycerol), Sulfuric acid (carbonized bioglycerol-supported sulfuric acid (solid acid))
Solvents: Ethanol; 8 h, 80 °C

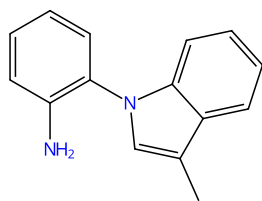
Experimental Protocols

By: Reddy, G. Niranjan; et al

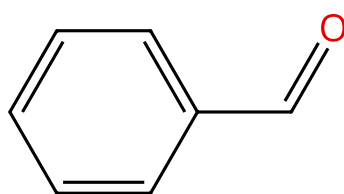
Canadian Journal of Chemistry (2015), 93(3), 341-347.

Scheme 46 (1 Reaction)

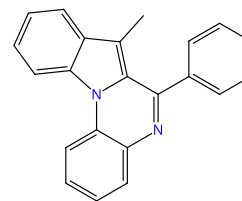
Steps: 1 Yield: 95%



+



→



Suppliers (2)

Suppliers (69)

Supplier (1)

31-087-CAS-4476452

Steps: 1 Yield: 95%

Lewis Acid-Catalyzed Selective Synthesis of Diversely Substituted Indolo- and Pyrrolo[1,2-a]quinoxalines and Quinoxalinones by Modified Pictet-Spengler Reaction

By: Verma, Akhilesh K.; et al

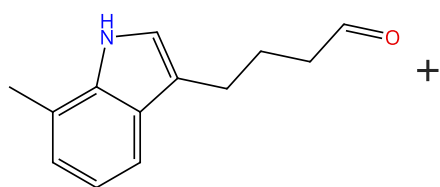
European Journal of Organic Chemistry (2011), 2011(34), 6998-7010, S6998/1-S6998/109.

1.1 **Reagents:** 1*H*-Benzotriazole, Oxygen
Catalysts: Aluminum chloride
Solvents: Tetrahydrofuran; 8 - 10 h, rt

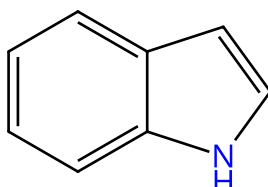
Experimental Protocols

Scheme 47 (1 Reaction)

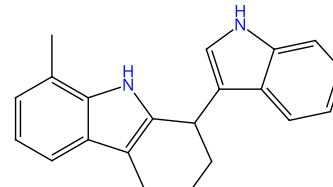
Steps: 1 Yield: 94%



+



→



Suppliers (113)

31-086-CAS-16361598

Steps: 1 Yield: 94%

Catalytic Enantioselective Synthesis of Tetrahydrocarbazoles and Exocyclic Pictet-Spengler-Type Reactions

By: Hansen, Casper L.; et al

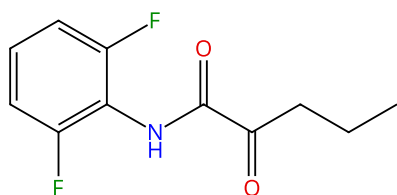
Organic Letters (2016), 18(23), 5990-5993.

1.1 **Catalysts:** Dinaphtho[2,1-*d*:1',2'-*f*][1,3,2]dioxaphosphepin, 4-hydroxy-2,6-bis[2,4,6-tris(1-methylethyl)phenyl]-, 4-oxide, (11*b**R*)-
Solvents: Dichloromethane; 5 h, -50 °C

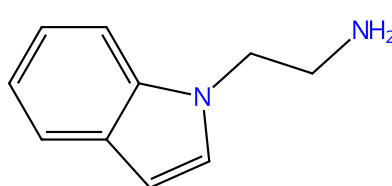
Experimental Protocols

Scheme 48 (1 Reaction)

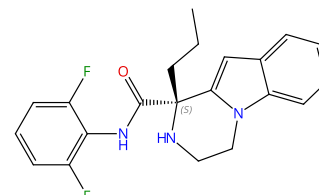
Steps: 1 Yield: 94%



+



→



Absolute stereochemistry shown,
Rotation (+)

Suppliers (48)

31-087-CAS-3755243

Steps: 1 Yield: 94%

Direct and highly enantioselective iso-Pictet-Spengler reactions with α-ketoamides: Access to underexplored indole core structures

By: Schonherr, Heike; et al

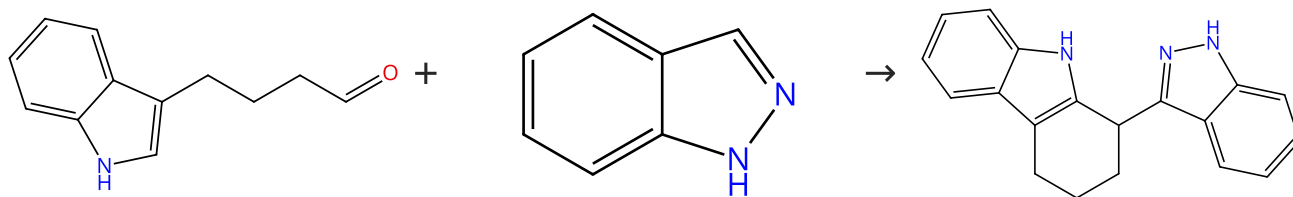
Organic Letters (2012), 14(10), 2610-2613.

1.1 **Solvents:** Toluene; 16 h, reflux; reflux → rt
 1.2 **Reagents:** (4*S*,5*S*)-2-Chloro-3,4-dimethyl-2,5-diphenyl-1-oxa-3-aza-2-silacyclopentane
Solvents: Toluene; 8 h, reflux; reflux → rt
 1.3 **Reagents:** Water; 30 min, rt

Experimental Protocols

Scheme 49 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (17)

Suppliers (94)

31-086-CAS-16361601

Steps: 1 Yield: 94%

Catalytic Enantioselective Synthesis of Tetrahydrocarbazoles and Exocyclic Pictet-Spengler-Type Reactions

By: Hansen, Casper L.; et al

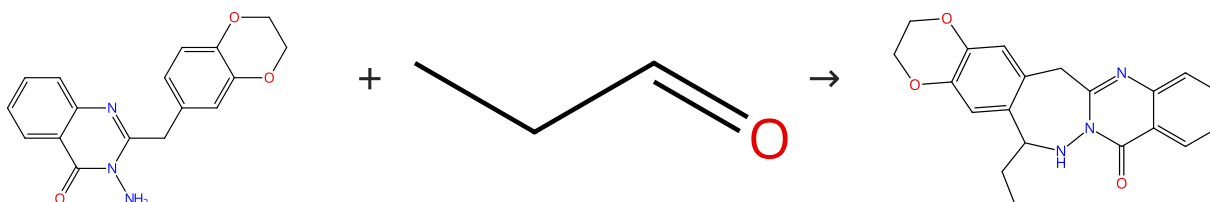
Organic Letters (2016), 18(23), 5990-5993.

1.1 **Catalysts:** Dinaphtho[2,1-*d*:1',2'-*f*][1,3,2]dioxaphosphepin, 4-hydroxy-2,6-bis[2,4,6-tris(1-methylethyl)phenyl]-, 4-oxide, (11*b* *R*)-
Solvents: Dichloromethane; 5 h, -50 °C

Experimental Protocols

Scheme 50 (1 Reaction)

Steps: 1 Yield: 94%



Suppliers (75)

31-087-CAS-10745384

Steps: 1 Yield: 94%

Pictet-Spengler reaction in the synthesis of condensed benzodiazepines. Part 2. Synthesis of new derivatives of 11, 12-dihydroquinazoline[3,2-*c*][2,3]benzodiazepin-14(6*H*)-ones

By: Tolkunov, A. S.; et al

Chemistry of Heterocyclic Compounds (New York, NY, United States) (2010), 46(6), 711-720.

1.1 **Reagents:** Hydrochloric acid
Solvents: 1,4-Dioxane, Water; 1 h, 80 - 90 °C; 80 °C → rt

1.2 **Reagents:** Ammonium hydroxide
Solvents: Water; 1 h, pH 4 - 5, rt

Experimental Protocols